MATH 314 - EXAM 2 STUDY GUIDE

Lecture 5 – Sampling Distributions

- Point Estimators ($\bar{X}$) (p.11-17)
  - Standard error (SD of a pt. est.)
  - Variance and standard error of $\bar{X}$
- Properties of Point Estimators (p.19-24)
  - Unbiased
  - Minimum Variance
- Sampling Distribution (p.26)
  - This is the distribution of $\bar{X}$, which is found by using the…
- Central Limit Theorem (p.28-29)
  - $\bar{X} \sim N(\mu, \sigma^2/n)$
  - We can **standardize** an estimator by subtracting by the mean and dividing by the standard error. Thus, $\left(\frac{\bar{X}-\mu}{\sigma/\sqrt{n}}\right) \sim N(0,1)$
  - The resulting standardized estimator will then approximately following the **standard Normal distribution**, N(0,1).

Lecture 5 Examples

- Using R to generate data for the CLT (p.3-7)
  - **Assignment 5**
- Using R to calculate bias (p.8-10)

Lecture 6 – Intro to Inference (When σ is known)

- Confidence Intervals (p.6-16)
  - Basic structure (estimator ± margin of error)
  - Defining the critical value $z^*_{1-\alpha/2}$ with qnorm(1-α/2)
  - ME = $z^*_{1-\alpha/2}$ x $\sigma/\sqrt{n}$
  - Interpretation (We are 100(1-α)% certain that the true population mean or proportion lies in this interval.)
- Hypothesis Tests (p.18-29)
  - Testing Framework
    - Define hypotheses
    - Collect data
    - Analyze data
    - Conclusions
  - Defining the null and alternative hypotheses
    - (Example: H_0: $\mu_0 = 5$; H_1: $\mu_0 \neq 5$)
    - Remember, the null hypothesis H_0 always uses "="
  - Define relevant variables ($\bar{X}, \sigma, n, \alpha$)
  - Defining the test statistic, $z = \frac{\bar{X}-\mu_0}{\sigma/\sqrt{n}}$, where μ_0 is the null hypothesis value
  - Defining and finding the p-value (using pnorm)
  - Comparisons to the confidence interval
  - **Assignment 6, problem 1**

Lecture 6 Examples

- One-tailed Tests and Confidence Intervals (p.7-12)
  - Upper vs. lower confidence intervals
  - Changes for proportion estimates
  - **Assignment 6, problem 2**

Lecture 7 – t-Distribution (When σ is NOT known)

- t-Distribution definition (p.8-13)
  - Comparison to standard normal distribution
  - Degrees of freedom (n-1)
  - R functions (qt, pt, and rt)
- Confidence Intervals (with t-distribution) (p.15-18)
  - Similar structure, with a few changes:
    - Use qt instead of qnorm
    - Use s (sample standard deviation) instead of σ
  - **Assignment 7, problem 2**
- Hypothesis Tests (with t-distribution) (p.19-24)
  - Similar structure, with one change:
    - Use $s_{\bar{X}}$ instead of $\sigma_{\bar{X}}$
  - Test statistic: $t_{df} = \frac{\bar{X}-\mu_0}{s/\sqrt{n}}$
  - For p-value, use pt instead of pnorm

Lecture 7 Examples

- Using R for plots and extracting variables from a data set. (p. 4-15)
  - **Assignment 7, problem 1**

Lecture 8 – t-Tests (Paired and Two-Sample Cases)

- One Sample t-test (review) (p.6)
- Two-Sample Paired Data (p.7-18)
  - Definition
  - The estimator, $\bar{X}_{diff}$, is the mean difference between each PAIR of data (NOT the difference between the overall group means)
  - Create a new variable containing the differences between each pair, then use it to perform a one-sample t-test
  - **Homework 8, problem 2**
- Two Sample Independent Data (p.20-33)
  - The estimator, $\bar{X}_a - \bar{X}_b$, is the difference between the two samples' overall means.
  - Standard error: $\sqrt{\frac{s_a^2}{n_a} + \frac{s_b^2}{n_b}}$
  - Test statistic: $t_{df} = \frac{(\bar{X}_a-\bar{X}_b)-(\mu_a-\mu_b)}{\sqrt{\frac{s_a^2}{n_a}+\frac{s_b^2}{n_b}}}$
  - Confidence interval: $(\bar{X}_a - \bar{X}_b) \pm t^*{}_{1-\alpha/2,df} * \sqrt{\frac{s_a^2}{n_a} + \frac{s_b^2}{n_b}}$
  - Degrees of freedom: $df = \min{(n_a - 1, n_b - 1)}$
  - **Assignment 8, problem 1**